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Taneem Ullah Jan

Applied AI researcher and engineer building production-grade AI systems across LLMs, retrieval, agentic workflows, and evaluation infrastructure: from research to reliable real-world deployment.

Professional Experience

AI Lead — Oct. 2025 – Present

ErisAI

- Architect and develop **Synapse**, a research-driven AI infrastructure layer for multi-agent workflows, integrating hybrid retrieval, multi-turn memory, MCP-based tool orchestration, intelligent model routing, and evaluation and observability protocols using Ragas and LangFuse to catch traceability and knowledge-access failures. (Synapse)
- Deployed **Synapse**, incorporating a LoRA fine-tuned model via Unsloth, across multiple enterprise AI applications including legal and investigative platforms, processing over 30M+ private documents with grounded retrieval and traceable responses. (Use case)
- Built **Ownify AI**, a multi-tenant white-label platform on top of Synapse with structural tenant isolation, per-tenant billing and usage metering, and workspace branding for organizations running their own AI product. (Product)

AI Researcher — Sep. 2024 – Oct. 2025

VOLV AI

- Researched and developed a 3D virtual try-on SDK using neural rendering, GANs, and diffusion models; the system was adopted by fashion brands and increased user engagement by 40%. (Demo)
- Developed controllable 2D and 3D digital human systems with pose estimation and generative modeling, supporting personalized experiences and reducing product return rates by 30%.
- Prototyped multimodal LLM systems integrating language, vision, and audio for cross-modal reasoning and instruction-following in customer support and interactive applications.

Research AI Engineer — Jan. 2023 – Dec. 2023

BHuman AI

- Built scalable generative video pipelines using neural reenactment, facial motion transfer, and voice cloning, reducing production costs by 40% for a platform serving 90K+ users and enterprise media clients. (Use cases)
- Researched and deployed audio-driven lip-sync models using GANs and super-resolution, improving visual fidelity and establishing a core product capability and primary revenue stream.
- Integrated LLMs with persona-driven avatars and developed retrieval-based conversational systems for enterprise clients, combining private knowledge, persona conditioning, and response evaluation for real-world deployments.

Intern Machine Learning Engineer — Aug. 2021 – Nov. 2021

NAECO Blue GmbH [*Profile*]

- Developed machine learning models to improve the spatial and temporal resolution of weather data for solar and wind energy forecasting.
- Built a data pipeline that reduced R&D time by nearly 50% and supported forecasting workflows later adopted by 5+ European meteorological agencies.

Selected Projects

Synapse (2026). Research-driven AI system integrating retrieval, memory, routing, evaluation, and runtime controls for reliable model behavior.

Ownify AI (2026). Multi-tenant white-label AI platform built on Synapse, with structural tenant isolation, per-tenant billing, and operational tooling for organizations running their own AI product.

PatchProof (2026). Framework for evidence-grounded patch verification using execution traces, behavioral validation, and correctness analysis beyond unit tests.

DarwinPatch (2026). Verifier-grounded coding-agent repair framework combining bounded evidence, route-aware search, and regression gates, lifting solve rate from 0.6 to 0.9.

GateGRPO (2026). Verifier-grounded reinforcement learning (GRPO) for code repair, replacing a learned reward model with the hard-gate verifier itself, beating SFT on held-out solve rate at lower search cost.

WarmLead (2026). Cache-liveness scheduler for lead-plus-sidekick LLM agent harnesses, with a cost simulator and live API verification to reduce idle-period token spend.

DGM-LLM (2025). Autonomous code self-improvement framework combining evolutionary search with LLM-guided mutation and multi-dimensional evaluation.

EmbedVoiceLLM (2025). Efficient multimodal framework mapping speech representations directly into LLM embedding space for low-latency voice interaction.

Skills

- **Programming**: Python, C++, Bash, MATLAB
- **LLM Systems**: RAG, Agentic Workflows, Custom Agent Orchestration, MCP (Model Context Protocol), Function Calling & Tool Use, Conversation Memory, Prompt Engineering (few-shot, structured output, system-prompt design), Prompt Caching & Context Window Optimization, vLLM, llama.cpp
- **Retrieval & Grounding**: Hybrid Retrieval, Vector Search, Cross-Encoder Reranking, Hallucination Detection (NLI Verification), Structure-Aware Document Chunking, Zero-Shot Information Extraction, Ragas, LangFuse
- **Vector Databases**: Qdrant, FAISS, pgvector
- **Frameworks**: LangChain, Haystack, FastAPI, Unsloth, TRL, OpenAI API, Anthropic API, Gemini API
- **ML Fundamentals**: PyTorch, Transformers, TensorFlow, NumPy, multimodal learning, LoRA, Quantization, SFT, Reinforcement Learning (RL, GRPO), Synthetic Data Generation, model evaluation, experiment design
- **Infrastructure**: Docker, Redis, Git, Linux, CI/CD (GitHub Actions), Systemd, Weights & Biases, Prometheus, Grafana, AWS, Azure, GCP
- **Computer Vision / 3D**: OpenCV, PyTorch3D, MediaPipe, ONNX

Publications

Taneem, U. J., & Ayesha, N. (2024). Beyond CNNs: Encoded Context for Image Inpainting with LSTMs and Pixel CNNs. International Journal of Innovations in Science & Technology, (IJIST) 6(5), 165–179, Special Issue ICTIS 2024.

Taneem, U. J., & Zakira, I. (2022). HTML Code Generation from Images with Deep Neural Networks. Journal of Engineering and Applied Sciences (JEAS), UET Peshawar. **(Young Undergraduate Researcher Award)**

Education

University of Engineering and Technology Peshawar, Pakistan

Bachelor of Science in Computer Science, CGPA: 3.58/4.0 — Sep. 2018 – Sep. 2022